

2020 AMC 10B Problem 9 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 9. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 9

Problem:

How many ordered pairs of integers (x, y) satisfy the equation

$$x^{2020} + y^2 = 2y?$$

Answer Choices:

- A. 1
- B. 2
- C. 3
- D. 4
- E. infinitely many

Solution:

Rearranging the terms and completing the square for y yields the result

$$x^{2020} + (y - 1)^2 = 1$$

Then, notice that x can only be 0, 1 and -1 because any value of x^{2020} that is greater than 1 will cause the term $(y - 1)^2$ to be less than 0, which is impossible as y must be real. Therefore, plugging in the above values for x gives the ordered pairs $(0, 0)$, $(1, 1)$, $(-1, 1)$, and $(0, 2)$ gives a total of (D) 4 ordered pairs.

OR

Bringing all of the terms to the LHS, we see a quadratic equation

$$y^2 - 2y + x^{2020} = 0$$

in terms of y . Applying the quadratic formula, we get

$$y = \frac{2 \pm \sqrt{4 - 4 \cdot 1 \cdot x^{2020}}}{2} = \frac{2 \pm \sqrt{4(1 - x^{2020})}}{2}$$

In order for y to be real, which it must be given the stipulation that we are seeking integral answers, we know that the discriminant,

$$4(1 - x^{2020})$$

must be non-negative. Therefore,

$$4(1 - x^{2020}) \geq 0 \implies x^{2020} \leq 1$$

Here, we see that we must split the inequality into a compound, resulting in $-1 \leq x \leq 1$. The only integers that satisfy this are $x \in \{-1, 0, 1\}$. Plugging these values back into the quadratic equation, we see that $x = \{-1, 1\}$ both produce a discriminant of 0, meaning that there is only 1 solution for y . If $x = \{0\}$, then the discriminant is nonzero, therefore resulting in two solutions for y .

Thus, the answer is $2 \cdot 1 + 1 \cdot 2 = \text{(D)} \boxed{4}$.

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